

Take home Midterm Problems and HW 4 Assignment 7502 Prof. Justin Frantz
Finalized Version – additions in navy blue and mostly in bold font
Due Thursday March 26, 2026

Part 1 Take-Home Midterm Problems. Please work on these independently, although you can always ask help from Prof. Frantz -- FINALIZED

M1) In a collider experiment, a decay of a certain heavy baryon can be identified from the decay to two particles, π^- and a proton as these particles are seen to emanate from a displaced vertex in a tracking detector (vertex located displaced from the beam/target locations). In a particular decay, the momenta of the π^- and p are measured to be 0.75 GeV and 4.25 GeV respectively, and the opening angle between the tracks is 9 deg. Use the masses of the pion and proton of 139.6 MeV and 938.3 MeV.

- a) Calculate the mass of the baryon.
- b) On average, over many such observed decays with the same total parent momentum, the displaced vertex is on average 0.35m from the point of production. Calculate the lifetime of the baryon.

M2) Two beams collide in an asymmetric energy collider ie a particle a in one beam and a particle b in the other have different energies. The collider intersects the beams such that (meaning the actual collisions occur when this is true) the standard convention that both beam a and b can be taken to be moving along the z -axis. In both parts below, the hyperbolic function relations involving y may be helpful –for example and also refer to rapidity problem **E1 (not 4)** from HW3. **[note that these are beam particles, so they obey the “alternative” hyperbolic relations mentioned at beginning of problem M4) below]**

- a) Show that the rapidity of the center of mass frame y_{CM} can be calculated by finding the rapidity of the four vector $(p_a^\mu + p_b^\mu)$. One way to do this is write the relations to explicitly transform p_z for a and b to the center of mass, and impose the required relation between p_a and p_b there.
- b) Use this result to show that if a and b are same mass particles, that $y_{CM} = \frac{1}{2}(y_a + y_b)$.
- c) Also use the basic transformation property of y given on the slides (first derived relation/property there) to show that in the CM frame the two particles have rapidities $\pm \frac{1}{2}(y_a - y_b)$
- d) Suppose we wish to use sPHENIX Central Arm “main detectors” (usual acceptance $-1 < \eta < 1$, in the CM frame with collisions positioned at the center) to study $p+p$ physics at rapidities farther away from mid-rapidity [$y=0$]. Thus RHIC is run with new asymmetric beam momenta of +9 GeV (the minimum “booster” feeder proton energy to the RHIC ring) and -250 GeV (the max proton beam energy at RHIC). Find the new CM energy ($\sqrt{s}$), y_a , and y_b . If the collisions are positioned to always happen at either end of the sPHENIX barrel, translate the resulting new η acceptance limits (see below) to what they would correspond to in the CM frame of reference. For this last task, use the formula in part b), think about c) conceptually, and, lastly, assume the $\Delta\eta$ size of the acceptance of sPHENIX always remains the same ($=2$), and, thus because of the moved collision point, we are actually just shifting the previous η acceptance to go now between 0 and ± 2 —make sure you justify why in this case by considering η ($=y$!) of detected particles on the same end of the barrel as the collision point. These acceptance limits, at least for the purposes of this problem, are not dependent on the beam energies, only on the collision point position, and thus stay $[-2,0]$ & $[0,+2]$ in the new asymmetric situation. Your job is to simply translate these limits into the corresponding CM y/η values. NOTE: This long winded looking additional question part should actually be quite easy, if not ask Prof. Frantz what you’re doing wrong.

(DRAWING the situation --formulas in b) vs. c)-- will also be very helpful)

M3) Similar to the example in Griffiths regarding the ratio of cross sections for $N+N \rightarrow d + \pi$, use isospin eigenstate kets to motivate why it is expected that the total cross section ratios for σ_{p+p} , σ_{p+n} , & (and σ_{n+n}) (e.g. pp/np) are all one, ie all cross sections should be considered equal for elastic scattering in some energy regime, like high but not too high energy.

M4) Short answers on rapidity: answer each of the questions below with at most two sentences. (each part a-through e may contain more than one question, the limit goes by number of questions, not parts).

a) In all cases in this problem when referring to rapidity y , the definition given in the class slides is meant, rather than e.g. the one defined by formulas connecting the boost factors to hyperbolic trig functions such as $y = \cosh^{-1}$ found in many books such as Mann (problem #2, 2.6 pg ~41) or L&B (~Eq 9.43). What's the difference between our definition and that textbook definition? (Note, the Thomson other alt textbook, Modern Particle Physics, does include a short treatment on the rapidity definition used in my slides, in section 10.9.1).

b) For practical purposes (in most cases where they are used) the numerical values of either pseudorapidity η or rapidity y , compared for the same particle are best described as which of the following: very different, different, similar, effectively the same, almost exactly the same, or exactly the same—include brief justification of this, though you may simply refer to information given in class. Which is preferred when considering when and if a maximum value should be considered?

c) Give two advantages of using y or η rather than the polar angle θ . Hint: relate at least one to the distribution of particles as a function of any of the three variables. For practical purposes, do these advantages apply to both y and η , or just one or the other?

d) Explain the reasoning behind saying that, and to what extent is it actually true, that the scale/size/"units" of the either y or η should be considered as radians? Include in your answer reference to the difference in rapidity between two particles, (or equivalently, between two directions, or two reference frames).

e) Why is the use of (pseudo) rapidity especially advantageous for high energy particle collisions involving at least one hadron or two ($X+h$ or $h+h$). Hint consider the so-called "parton distribution functions" pdf(x) representing the probability that a parton inside a hadron carries a fraction x , which randomly fluctuates event by event, of its parent hadron's 4-momentum.

Part 2 Regular Homework 4 Portion – This part feel free to collaborate, in addition to asking Prof. Frantz for help. FINALIZED

H1) Using the expression for $d\sigma/dt$ for $1+2 \rightarrow 3+4$ given in the slides [from Thomson], (and using our natural units throughout)

a) Express the formula using only Mandelstam variables if $m_1 = m_2 = m$. (& other constants but for all kine factors... ie instead of $|p_i^*|$ or $|p_f^*|$)

b) [somewhat like Griffiths 6.13 or 6.14 but for any frame] use it and the Feynman rules for the ABC theory, to derive the cross section $d\sigma/dt$ for $C+C \rightarrow A+A$ scattering to lowest order, also assuming propagating particle mass of 0, and the other two particle types have the same non-zero mass m , of course writing your final expression in terms of only Mandelstam variables ($=f(s,t,u)$). You may of course follow the similar short derivation in the book even go fairly straight to the results [note, for slightly different particles used], or go from commentary on that in the slides, but at least start by writing the diagrams and at least briefly justify why the ABC Feynman rules mean results wouldn't change essentially.

H2) Show that $\bar{u}(p_1)\gamma^\mu u(p_1) = K p_1^\mu$ where K is a constant. **Don't just use the result of H3), but rather show it explicitly using the forms of the $u / \bar{u}$ solutions given in Griffiths. We will probably do this in class together.**

H3) Partially Based on Peskin Problem 3.2

The two QFT reference books (Peskin et al/Lancaster et al), probably most QFT books, and some particle physics books apparently prefer a different ("alt") convention than the one Griffiths uses (apparently NOT Bjorken and Drell's per Griffiths) for the γ matrices explicit forms/representation. The difference is simply that the "alt" convention (apparently also called the Chiral or Weyl convention) "switch" γ^0 and $-\gamma^5$. Meaning the negative of the traceless matrix Griffiths calls γ^5 is the one they call γ^0 and the one Griffiths calls γ^0 they call γ^5 . This allows for amongst other things a more compact notation since e.g. then all components of γ^μ including $\mu = 0$ have 2x2 block form with zero $[0_{2 \times 2}]$ as the diagonal blocks and off-diags written in terms of $\sigma^\mu = (1_{2 \times 2}, \vec{\sigma})$.

a) Compare the explicit form (in terms of p^μ) of the matrix $\gamma^\mu p_\mu = \not{p}$ between the two conventions.

b) Compare the explicit form of the matrix $\sigma^{\mu\nu}$ defined in equation 7.69 in Griffiths for $\mu = 0$ and $\nu = 1$ between the two conventions.

c) [Peskin 3.2] Prove the "Gordon Identity":

$$\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p')\left(\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p).$$

Start from the RHS and simplify till you obtain the LHS using anti/commutation relations, the Dirac equation and possibly γ matrix identities given in Griffiths such as those in the previous problem (Giff problem 7.32). You may presume these all still apply as given in Griffiths, even in the case of the alt convention, for the purposes of this problem. For sure the anti-commutator $= 2g^{\mu\nu}$ relation MUST still hold for the Dirac trick of factorizing the Klein-Gordon equation to work.

PS.: In my solution, the explicit forms of the γ matrices were not needed, and therefore which convention shouldn't matter for this part of the problem. But just in case I had an error, if you get stuck you may want to verify whether any of the Griffiths relations you used still apply exactly as given when using the alt convention-- this is not required, only a suggestion if you run into trouble. Also, note, in the alt convention, the free particle solution $u(p)$ choice is also generally defined differently and/or there may be a slightly different normalization for $u(p)$ in the alt choice (or even perhaps just because of

a further choice of norm particular to Peskin & Schroeder) but try not to use the explicit forms for the $u(p)$'s to do this problem, you shouldn't need them, only that $\psi = u(p)e^{ip \cdot x}$ is the form of the solution to the Dirac equation. In case normalization conventions are different, you need only prove the relation up to an overall constant.

H4) Do parts of Griffiths 7.37. Obtain only the result given by the first expression for $\langle M_1 M_2^* \rangle$. Use the contraction theorems to evaluate the Trace as instructed but ONLY for that for that expression. Instead of adding to *that* the analogous expressions for the other 3 terms in the squared matrix element, simply use the given final answer expression, rewritten in terms of Mandelstam s, t & u to give an expression for the cross section, using the $d\sigma/dt$ formula. Note we are always ignoring the electron mass. Note we will give help on this problem in class.